

SEAT No. _____

No. of Printed Pages 2

[697A47]

SARDAR PATEL UNIVERSITY

B.C.A Semester - 4 (Reg. & NC)(CBCS)

Friday, Date: 7th April 2017

Session: Evening Time: 02:00 P.M. to 5:00 P.M.

Subject Code: US04CBCA03

Subject Title : OPERATING SYSTEMS

Total Marks: 70

Q1. Multiple Choice Questions. (Attempt all)

[10]

1. Which of the following is **NOT** a valid process state?
A. New C. Create
B. Ready D. Waiting
2. _____ scheduling algorithm gives minimal average waiting time.
A. Round Robin C. SJF
B. Priority D. FCFS
4. _____ scheduling algorithm gives minimum Page Faults.
A. FIFO C. LRU
B. SJF D. Optimal
5. In Best-fit memory allocation technique memory manager allocates _____.
A. First hole that is big enough to store process C. Smallest hole that is big enough to store process
B. Biggest hole that is big enough to store process D. Last hole that is big enough to store process
6. In Clock page replacement algorithm, replace the page with the value of R bit = _____.
A. 0 C. 1
B. i D. v
7. Each process has a segment of code called _____ section.
A. critical C. important
B. available D. mutual
8. Which of the following is **NOT** the valid resource utilization state?
A. Request C. Process
B. Use D. Release
9. _____ buffer has fixed buffer size.
A. UnBounded C. Flexible
B. Bounded D. Fixed
10. Printer is an example of _____ resource.
A. Physical C. Logical
B. Virtual D. Critical

(PTO)

- Q2. Answer the following short questions (Attempt any TEN) [20]**
1. Draw the diagram of PCB.
 2. What are Turnaround Time and Response Time?
 3. What is Preemptive Scheduling ?
 4. List different memory allocation techniques.
 5. What is Belady's Anomaly?
 6. Explain how to solve problem of External fragmentation.
 7. Explain when does a Race condition arise?
 8. Explain why is LINUX an open source?
 9. Justify "Linux is a Secure Operating System".
 10. Explain who command in brief.
 11. Explain man command in brief.
 12. Explain cp command with all its attributes.
- Q3. Explain SJF scheduling algorithm in detail. [10]**
- OR**
- Q3. Explain Round-Robin scheduling algorithm with example. [10]**
- Q4.a. List out different page Replacement algorithms and Explain Optimal Page Replacement Algorithm in detail. [06]**
- b. Define Page Fault. Explain page fault handling technique. [04]**
- OR**
- Q4.a. Explain FIFO page replacement algorithm with advantages. [06]**
- b. What is Fragmentation? List different types of Fragmentation. Explain any one in brief. [04]**
- Q5.a. Explain the concept of Process Synchronization in detail. [05]**
- b. What is LINUX? Explain basic features of LINUX Operating System. [05]**
- OR**
- Q5.a. What do you mean by Deadlock? Explain all necessary conditions for occurrence of deadlock. [05]**
- b. What is a Cooperative Process? Explain Producer-Consumer Problem in brief. [05]**
- Q6.a. Explain if and case statement in LINUX. [05]**
- b. Explain following commands: i) pwd ii) rmdir [05]**
- OR**
- Q6.a. Explain grep command with at least four possible attributes and examples. [05]**
- b. Explain following commands: i) ls ii) mkdir [05]**
- X-----